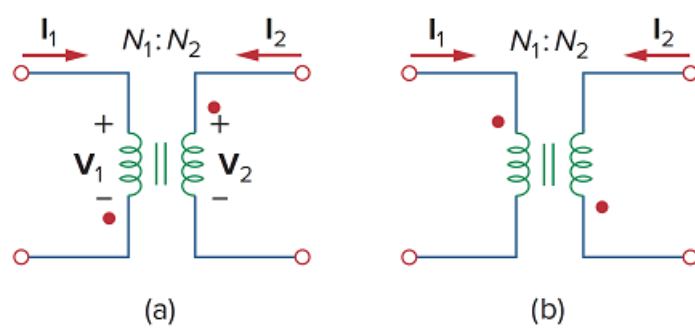


Problem

The ideal transformer in Fig. 13.70(a) has $N_2/N_1 = 10$. The ratio V_2/V_1 is:

- (a) 10
- (b) 0.1
- (c) -0.1
- (d) -10

Figure 13.70



Step-by-step solution

Step 1 of 2

Refer to Figure 13.70 (a) in the textbook.

Consider the relation between turn's ratio and transformer voltages.

$$\frac{N_2}{N_1} = \frac{V_s}{V_p} \dots\dots (1)$$

Where

V_s is the secondary voltage

V_p is the primary voltage

Step 2 of 2

From the Figure,

The primary voltage of the transformer is,

$$V_p = V_1$$

The secondary voltage of the transformer is,

$$V_s = -V_2$$

Substitute $-V_2$ for V_s , V_1 for V_p and 10 for $\frac{N_2}{N_1}$ in equation (1).

$$10 = \frac{-V_2}{V_1}$$

$$\frac{V_2}{V_1} = -10$$

Thus, the correct answer is **d**.